DermAI

Compliance Report

Supervisor

Dr. Taha Shabbir

Co-supervisor

Mr. Aamir Hussain

Submitted by

Huzaifa Bin Shakeel Shaikh (2539-2022)

Shahzaib Hussain Pirzada (2541-2022)

Abdul Wasay (2694-2022)

Department of Computer Science,

Hamdard University, Karachi.

Presentation Date: 08/07/2025

Compliance Report Date: 25/07/2025

1. Introduction

This report confirms that our Final Year Project Proposal — DermAI — is in full compliance with the academic, technical, and documentation standards set by Hamdard University. It outlines the sections completed in our FYP-I proposal, details how we addressed each evaluation criterion, and reflects the revisions and improvements made after receiving jury feedback.

2. Objectives of the Report Submission

This report was submitted with the purpose to:

1) Propose a feasible and impactful AI-based healthcare solution
2) Define the project scope, architecture, and deliverables
3) Explain the methodology used (Waterfall Model)
4) Present well-structured planning (Gantt chart, milestones)
5) Highlight use of modern tools like, CoAtNet, Firebase, Flutter, and BERT

3. Report Requirements

As per departmental guidelines, our FYP-I report and presentation fulfilled the following expectations:

Requirement	Status	Explanation
Title Page and Abstract	✅ Met	Properly included with clear project summary
Problem Statement	✅ Met	Clearly defines the issue of self-diagnosis & lack of human consultation
Literature Review	✅ Met	Compared DermAI to Aysa, Google Dermatology Assist, SkinGPT-4, etc.
Project Scope & Objectives	✅ Met	Explained focus on 5–7 diseases, doctor consult, smart questioning
Methodology (Waterfall)	✅ Met	Clearly described: analysis → design → training → testing → deployment
Architecture Diagram	✅ Met	Big picture included (AI model, app, backend, doctor interface)
Workflow Breakdown	✅ Met	Roles divided via RACI chart; milestones and deliverables listed
Project Plan (Gantt Chart)	✅ Met	Weekly plan from Aug to June with clear phase divisions
Budgeting	✅ Met	PKR 50,000 breakdown:

		Firebase, APIs, devices, printing, QA
Tools and Technologies	✅ Met	Python, TensorFlow, Flutter, Firebase, BERT, GitHub listed
Software & Hardware Requirements	✅ Met	Minimum specs defined: 8GB RAM, Android phone, Google Colab
Project Deliverables	✅ Met	APK, trained model (.tflite), doctor consult module, report, and video demo

4. Compliance Assessment

The following table evaluates our project based on core compliance metrics:

Requirement	**Status**	**Explanation**
Project Scope	✅ Met	Focused on skin detection + live consultation + symptom questions
Architecture Diagram	✅ Met	Architecture slide included (mobile app, AI model, backend, Firebase)
Workflow & Methodology	✅ Met	Applied Waterfall; clearly divided into 6 steps with detailed breakdown
Gantt Chart Aligned with Timeline	✅ Met	48-week Gantt chart included from Aug to June
Jury Feedback Addressed	✅ Met	All feedback reflected in new version: tools justification, dataset clarity

5. Comparison

Feature / Model	EfficientNet	CoAtNet	ConvNeXt	Swin Transformer
Architecture Type	CNN (Compound Scaling)	Hybrid (CNN + Transformer)	Modernized CNN (Inspired by Transformer)	Pure Transformer (Shifted Window attention)
Core Strength	Lightweight & efficient	Combines local + global feature extraction	Transformer-level accuracy with CNN simplicity	Global attention & fine-grain image analysis
Accuracy	High (B7 ~84% top-1 on ImageNet)	Higher than EfficientNet on multiple benchmarks	Matches/outperforms ViT & EfficientNet	Very high on medical imaging tasks
Mobile Deployment	✅ Supported via TFLite	✅ Possible (with tuning)	✅ Suitable for mobile (slightly heavier)	❌ Heavier, needs optimization for mobile
Training Speed	Fast (pretrained available)	Medium (slightly more complex)	Medium	Slower (transformers require more resources)
Feature Extraction	Strong on shallow to mid features	Excellent local + global understanding	Deep pattern recognition with improved design	Very strong global & contextual recognition
Suitability for DermAI	✅ Proven and mobile-optimized	✅ Best for hybrid systems with LLM integration	✅ Powerful and easier to train than transformers	✅ Ideal if resources allow high-end training

For our project, CoAtNet is the best architecture to use because it integrates the benefits of CNNs and Transformers which allows for both local feature extraction and global attention. This kind of design improves accuracy in the diagnosis of complex patterns on the skin. Nevertheless, we are still flexible to change to other models based on performance metrics and hardware resources, focus on accuracy thresholds, and the needs of deployment.

1. EfficientNet

- 📄 **Paper:** https://arxiv.org/abs/1905.11946

2. CoAtNet

- 📄 **Paper:** https://arxiv.org/abs/2106.04803

3. ConvNeXt

- 📄 **Paper:** https://arxiv.org/abs/2201.03545

4. Swin Transformer

- 📄 **Paper:** https://arxiv.org/abs/2103.14030

6. Comparative Analysis

Feature / Criteria	**DermAI (Your Project)**	**Aysa (VisualDx)**	**Google Derma Analytica**
Core Technology	Yes – CNN model (EfficientNet/CoAtNet)	CNN-based visual classifier	Deep learning model
Image-Based Prediction	✅ Yes, using pre-trained EfficientNet /CoAtNet	✅ Yes	✅ Yes
Cross-Questioning (LLM)	✅ Yes – symptom-based dynamic follow-up questions	❌ No symptom questioning	❌ No interactive dialogue
Language Support	English (with potential for via mBERT)	English only	English (limited multilingual output, no

Feature / Criteria	**DermAI (Your Project)**	**Aysa (VisualDx)**	**Google Derma Analytica**
Doctor Consultation	✅ Live or asynchronous consultation via video/chat	❌ Not available	❌ Not offered
Target Skin Conditions	5–7 common non-critical conditions (eczema, fungal, etc.)	Broad range (based on VisualDx's clinical library)	288+ conditions (incl. rare/malignant ones)
Platform	Cross-platform: Desktop + Android	Mobile app (iOS/Android)	Web-based (browser)
Offline Availability	❌ Online only (cloud LLM + real-time consult)	✅ Some features offline (image diagnosis only)	❌ Requires Internet
User Interaction Level	✅ High – AI plus doctor plus LLM makes it interactive	❌ Low – static suggestion list only	❌ Low – image upload with result only
Privacy & Data Handling	GDPR-style security, patient consent required	HIPAA compliant	Google-managed anonymization

Google Dermatology Assist

https://health.google/consumers/search/

Aysa

- 🌐 **Official Website:** https://www.askaysa.com

8. Classifier Comparison

Feature / Aspect	**Softmax Classifier**	**BERT-Based Classifier**	**Logistic Classifier**	**Ensemble (Fusion) Classifier**
Input Type	Image (CNN features)	Text (user symptom answers)	Text or extracted binary features	Outputs from Softmax + BERT
Type	Multi-class classifier	Binary / Contextual classifier	Binary classifier	Hybrid / Rule-based ensemble
Purpose	Identify 1 out of multiple diseases	Understand presence of symptoms	Confirm specific features (e.g., redness)	Final decision-making by combining predictions
Model Used	EfficientNet / CoAtNet (CNN) + Softmax	BERT / mBERT	Logistic Regression	Weighted logic or learned fusion layer
How it Works	Outputs class probabilities	Interprets user input using language context	Classifies symptom as yes/no	Merges image and text outputs for final result
Strength	Fast, accurate visual classification	Deep understanding of symptom descriptions	Simple, interpretable for feature detection	Improves overall accuracy and user confidence
Limitation	Cannot understand text/symptoms	Can't process visual inputs	Limited scope per feature	Needs careful tuning and logic design

9. Methodology – Waterfall Model

Requirement Gathering and Analysis

- Identified the core need: an AI-based skin disease diagnostic tool with doctor consultation support.
- Defined functional and non-functional requirements.
- Conducted domain research and analyzed user expectations and ethical considerations.

System Design

- Chose architecture: hybrid system using CNN (EfficientNet/CoAtNet) + LLM (BERT/GPT).
- Decided tools and frameworks (TensorFlow, PyTorch, Twilio, Firebase).
- Designed data flow, UI wireframes, and model integration structure.

Implementation (Coding Phase)

- Developed the image classification model using transfer learning.
- Integrated BERT-based intelligent questioning module.
- Built frontend for both mobile and desktop using cross-platform tools (e.g., Flutter).

Testing

- Conducted unit testing on image classification accuracy.
- Validated LLM's question understanding and response evaluation.
- Tested integration of modules and database (Firebase), along with API interactions.

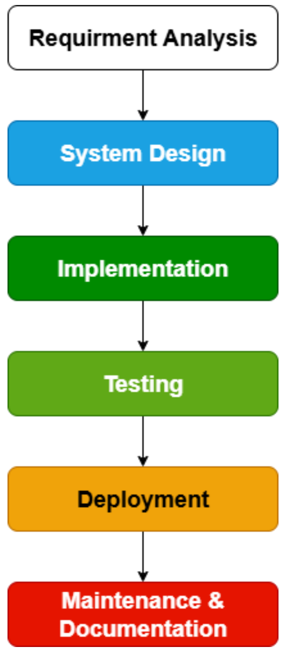
Deployment

- Deployed working prototype online.
- Converted model to mobile-friendly format (e.g., TFLite) for real-time testing.
- Enabled doctor consultation via Twilio and user authentication via Firebase.

Maintenance and Updates

- Incorporated jury feedback (model upgrade, classifier expansion).
- Planned for model retraining with additional user data.
- Designed a roadmap for multilingual support and ethical updates.

Waterfall was chosen for DermAI due to the project's well-defined scope, the medical domain's structured nature, and the need for sequential development from data acquisition to deployment.



10. Why We Use Transfer Learning

Less Data Needed: We don't have millions of images — so we reuse models trained on large datasets like ImageNet.

Saves Time & Resources: Pre-trained models already know general image features, reducing training time and compute cost.

Boosts Accuracy: Even with small medical datasets, transfer learning gives high performance using strong feature detectors.

Optimized for Mobile: Lightweight models like MobileNetV2 work well on mobile devices when converted to TensorFlow Lite (TFLite).

Faster Development: No need to build a model from scratch — we simply fine-tune it for our specific skin disease categories.

11. Changes Made After Jury Feedback

-Model Justification Improved: Compared multiple models (EfficientNet, CoAtNet, etc.) and selected CoAtNet for its hybrid performance.

-Classifier Section Added: Clearly described Softmax, BERT, Logistic, and Ensemble classifiers with their roles in DermAI.

-Existing Tools Compared: Compared DermAI with Aysa and Google Dermatology Assist in terms of features, interactivity, and language support.

-Tool Selection Explained: Justified why MobileNetV2 and EfficientNet were initially considered (lightweight, mobile-friendly).

-Dataset Sources Mentioned: Included HAM10000 (Kaggle) and ISIC Archive as image dataset sources.

-**Milestones Updated:** Adjusted development timeline; LLM integration moved earlier for better module alignment.

-Smart Questioning Clarified: Highlighted that GPT-powered follow-up questions are non-rule-based and dynamic.

-Waterfall Justification Added: Justified Waterfall model due to clear scope, medical accuracy needs, and structured development flow.

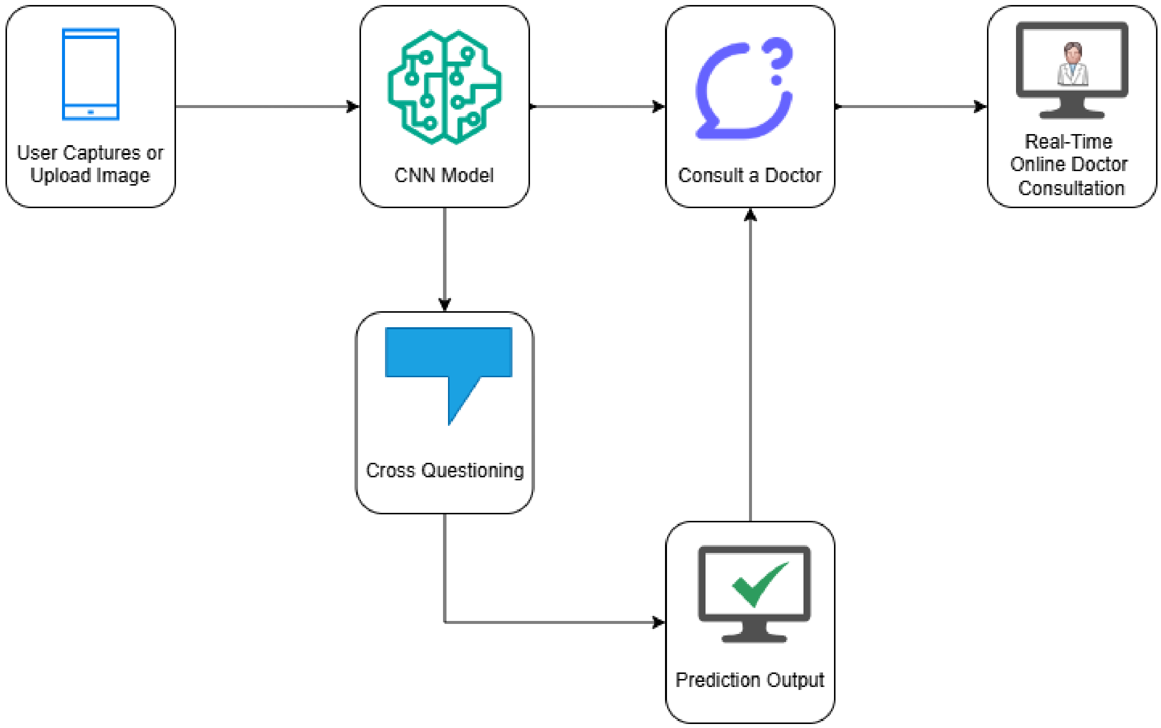
12. Challenges and Resolutions

Challenge	**Resolution**
Finding high-quality labeled datasets	Used HAM10000 and ISIC — open-source, medical-grade skin datasets
Smart questioning (non-rule-based implementation)	Integrated OpenAI GPT for dynamic question generation
Live doctor consultation integration	Used Firebase/Twilio simulation for video connection
Device testing limitations	Used Android phones with varying specs for testing; emulator for fallback

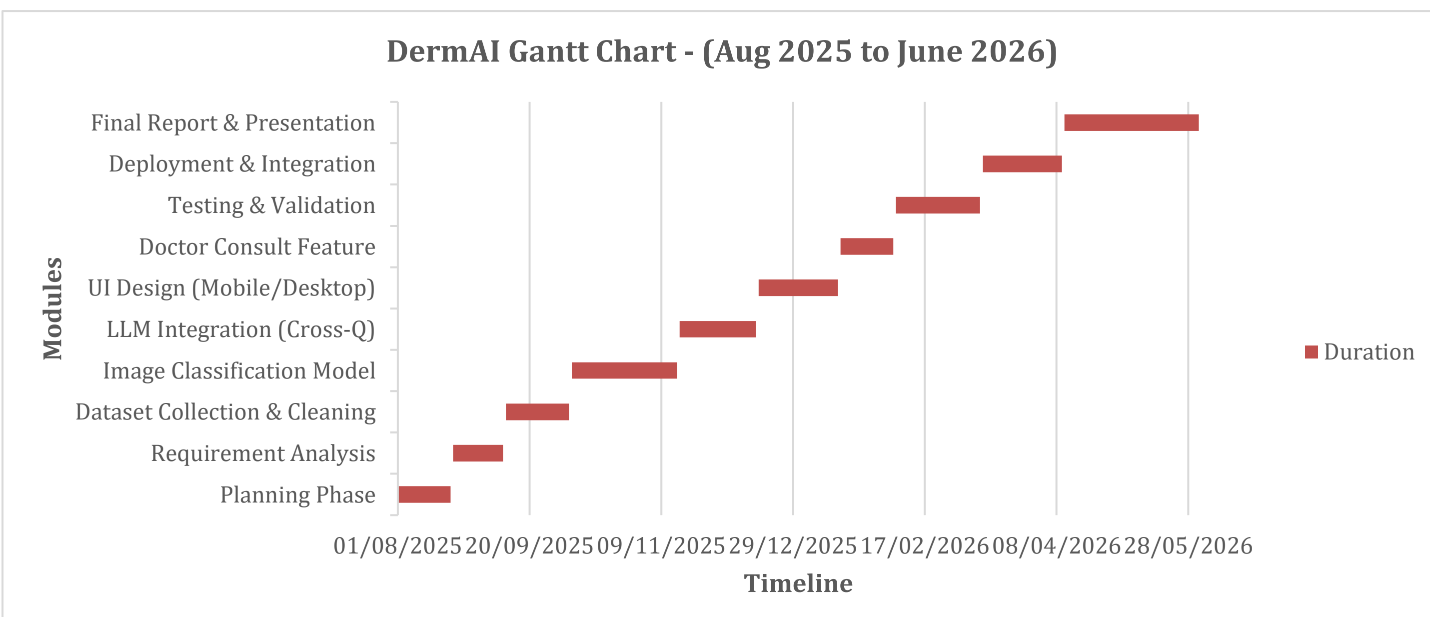
13. Supporting Evidence

1) Updated FYP-I Report document (post-feedback version) submitted with the compliance report.

2) System Architecture Diagram (AI model, app, Firebase, doctor)



3) Revised Gantt Chart



Project Timeline & Milestones

Milestone 1: Planning Phase

- Establish team structure and responsibilities
- Select project title, objectives, and feasibility
- Identify tools, technologies, and dataset sources

Milestone 2: Requirement Analysis

- Conduct background research on dermatology and AI
- Define system requirements and expected features
- Document functional and non-functional specifications

Milestone 3: Dataset Collection & Cleaning

- Acquire HAM10000 / ISIC datasets
- Remove irrelevant data and preprocess images
- Label and organize the dataset for model input

Milestone 4: Image Classification Model

- Train and evaluate CNN (EfficientNet or MobileNetV2)
- Tune hyperparameters for optimal performance
- Convert trained model to TFLite for mobile integration

Milestone 5: LLM Integration (Cross-Questioning Module)

- Implement symptom-based dynamic questioning flow
- Integrate LLM (e.g., BERT or GPT) for follow-up prompts
- Validate contextual accuracy of user inputs

Milestone 6: UI Design (Mobile/Desktop)

- Design clean, responsive UI for both mobile and desktop
- Develop core app functionalities (image upload, results display)
- Connect UI to AI modules and backend services

Milestone 7: Doctor Consultation Feature

- Implement live video or asynchronous doctor consultation
- Use platforms like Firebase, Twilio, or third-party APIs
- Ensure consultation data is stored securely and ethically

Milestone 8: Testing & Validation

- Conduct system, integration, and user testing
- Evaluate performance, usability, and accessibility
- Fix bugs and refine user experience

Milestone 9: Deployment & Integration

- Deploy web and mobile apps with full functionality
- Monitor system logs, usage data, and user feedback
- Ensure all modules work in unison

Milestone 10: Final Report & Presentation

- Prepare final documentation and user manual
- Record a demo video explaining features and workflow
- Submit and present the complete project

4) Jury Presentation Feedback (verbally acknowledged and addressed)

Project Name:	Score /40	Jury Comments
DermAI	25	Dataset usage not clear if using the Transfer Learning. Simply doing classification task.
		Literature Review required to explicitly present the idea. Background knowledge lagging
		Above average
		The literature review should more clearly convey the central concept, as it presently falls short in providing adequate background and context.
FYP Committee Comments:	Please update methodology and literature review.	
FYP Convener Recommendation:	**Conditionally Accepted** (Re-submit updated proposal document along with the compliance report by 25 July)	

14. Conclusion

This compliance report confirms that the revised DermAI proposal meets all academic and technical standards for the FYP-I phase. The documentation covers problem identification, realistic scope, technical feasibility, budget, and planning with clear visual aids. All suggestions from the jury were incorporated to enhance clarity, feasibility, and depth of the project.